europass

Romolo Di Bernardo

Date of birth: 24 Jul 1992 | **Place of birth:** Gaeta, Italy | **Nationality:** Italian | **Gender:** Male |

Phone number: (+39) 3299367450 (Mobile) | **Email address:** romolo.dibernardo@gmail.com |

Email address: romolo.dibernardo@unina.it | **LinkedIn:** Romolo Di Bernardo |

Address: via Garigliano 5, 04024, GAETA, Italy (Home)

ABOUT ME

Ph.D. Engineer specializing in aero/hydrodynamics, advanced design, and numerical methods, including parametric and algorithmic modeling. Skilled in managing the entire design cycle, executing numerical simulations, and conducting detailed data analysis. Currently a Postdoctoral Researcher at the University of Naples Federico II. Proven track record in hydrofoil design and tailored solutions for racing sailing yachts, with the ability to lead complex teams and make strategic technical and managerial decisions.

WORK EXPERIENCE

POST-DOC RESEARCH FELLOW – UNIVERSITY OF NAPLES FEDERICO II IN COLLABORATION WITH LEONARDO S.P.A. – Apr 2025 – Current – NAPOLI, ITALY

Generative and Biomimetics Design for Additive Manufacturing of Aeronautical Structures in the context of the NEMESI project (New Engineering and Manufacturing Enhanced System Innovation) *Leonardo S.p.A*.

- **Advanced Design for Aeronautical Components:** Engineered generative and biomimetic designs for **primary and secondary wing structures**, utilising Design for Additive Manufacturing (DfAM) to reduce mass while maintaining structural integrity.
- **Structural Optimisation:** Performed **topology and topography optimisation** to maximise stiffness-to-weight ratios on critical aerospace components.
- **Fluid-Structure Interaction (FSI):** Conducted FSI analyses to predict load scenarios on bio-inspired geometries under aerodynamic stress.
- **Experimental Hydrodynamics (LEIN Laboratory):** Collaborated with the Towing Tank Laboratory to investigating **hydrofoil dynamics and free-surface interaction phenomena** (ventilation and wave patterns).

INDEPENDENT ENGINEERING CONSULTANT – 2021 – Current

- **Dedalo Navi S.r.l.:** Contributed to the PNRR-funded "Greenta" (Balta S.r.l.**)** project for the development of a **high-speed electric hydrofoil**. Executed the design and optimisation of hulls and hydrodynamic appendages, performing fluid dynamics analyses to maximise efficiency and stability. Ongoing consultancy as fluid dynamics consultant.
- **NACRA 17 ITA98 - Ugolini/Giubilei:** Active technical partnership involving the redesign critical structural components to enhance functionality and **minimise aerodynamic/hydrodynamic drag**. Conducted iterative design cycles ensuring strict compliance with class rules and regulations.
- **Boat Propulsion Italia S.r.l:** Ongoing R&D collaboration focused on performing advanced CFD campaigns to analyse and optimise propulsion systems, including **open-water** and **ducted propellers,** analyses of planing hulls and complex flow regimes.
- **Racing Yacht Consultancy (F. Colaninno):** Applied **Reverse Engineering** and CFD solvers to optimise racing sailboats. Focused on identifying performance gains through precise geometric modifications of hull and appendages.
- **Design for Additive Manufacturing (DfAM):** Developed bespoke industrial applications for different companies, utilising topology optimisation to create **lightweight, high-strength structures** tailored for advanced manufacturing processes.
- **Italian Armed Forces (Guardia di Finanza):** Designed infrastructure systems, including **mooring systems** for the Naval School and **high-power charging stations** for military vessels, ensuring reliability under operational constraints.

TECHNICAL-SCIENTIFIC SUPPORT COLLABORATOR – SCIENTIFIC CONSORTIUM FOR THE INDUSTRIAL RESEARCH AND ENGINEERING (S.C.I.R.E.) – 1 Jun 2025 – 31 Dec 2025 – ROME, ITALY

- **VR Pipeline Development:** Developed a custom automated workflow utilising **Python scripting** and **Grasshopper** to interface with **Meta Quest VR** hardware, effectively bridging the gap between raw computational data and immersive analysis.
- **Big Data Integration:** Engineered robust data import pipelines to handle large-scale datasets, enabling the visualisation of complex simulation results within virtual environments.
- **Experimental Workflow Design:** defined new protocols for scientific observation in VR, enhancing data interpretability and speeding up the decision-making process for experimental campaigns.

CONTRACT TECHNICAL INSTRUCTOR – IN.SI. SRL (CLIENT: AVIO S.P.A.) – Jan 2025 – Feb 2025 – FERENTINO, ITALY

- **Delivered the specialised course** *"Topology and Structural Optimization Procedures"* to professional aerospace and aeronautical engineers.
- **Instructed on advanced design methodologies**, focusing on Generative Design and Design for Additive Manufacturing (DfAM) for high-performance components.
- **Covered structural validation workflows**, including Finite Element Analysis (FEA) advanced design methodologies and topology optimisation strategies for weight reduction.

ADJUNCT PROFESSOR – IPSEOA A. CELLETTI – Feb 2022 – Jun 2022 – FORMIA, ITALY

Course: Organization and Management Techniques for Production Processes.
Focus: Taught methodologies for industrial process optimization, production planning, and resource management, providing future engineers with tools to streamline manufacturing workflows.

CO-FOUNDER AND TECHNICAL PROJECT MANAGER – VISIO ELECTRIC S.R.L. – 2019 – Current – GAETA, ITALY

- **Directed technical strategy and business operations**, overseeing budgeting, resource allocation, and project delivery timelines.
- **Led and mentored the technical team**, fostering a high-performance culture and technical growth.
- **Spearheaded R&D and product development** for bespoke industrial solutions, managing the full lifecycle from concept to manufacturing.

EDUCATION AND TRAINING

2022 – 2024 Cassino, Italy
PHD IN INDUSTRIAL ENGINEERING: METHODS, MODELS, AND TECHNOLOGIES; 37TH CYCLE DOCTORATE. University of Cassino and Southern Lazio

- **Visiting Researcher at CDRSP (Politécnico de Leiria, Portugal)** | *Feb – Sept 2024* Developed **algorithmic and parametric design methodologies** for hydrofoils using **Rhinoceros** and **Grasshopper**. Focused on Rapid and Sustainable Product Development (CDRSP).
- **Experimental & Numerical Analysis (University of Naples Federico II)** Conducted integrated research at the **LEIN Towing Tank Laboratory** with numerical simulations (**CFD, FEM, and Fluid-Structure Interaction**).
- **Industrial R&D Placement – Rivale Yacht Srl** | *Jan – Oct 2022* Collaborated directly with **Valerio Rivellini Yacht Design** on the development of the **hybrid motor-yacht *Vertus***. Applied advanced engineering principles to real-world naval architecture constraints during the prototyping and production phases.

Website www.unicas.it |

Field of study Mechanical Engineering Design Methods, Fluid Dynamics applied to Hydrofoil Design, Algorithmic and Parametric Design, Numerical Analyses, Naval Engineering

Final grade with Honors | **Thesis** Development of Algorithmic and Parametric Methods for the Design of Sustainable Hydrofoil Boats

2022 – 2023 Cambridge, United States
HARVARDX'S COMPUTER SCIENCE 50 edX & HarvardX University

Covered **algorithms**, **data structures**, and **computational thinking**.
Gained proficiency in **C** and **Python** programming, applying code-based problem solving to complex scenarios.

Website https://courses.edx.org/certificates/87e9f91e9e46481b9880acaacf33f8b3 | **Final grade** Professional Certification Owner

Frosinone, Italy
LICENSED PROFESSIONAL ENGINEER (STATE EXAMINATION FOR PROFESSIONAL PRACTICE) Order of Engineers of Frosinone, Italy

Address Via Tommaso Landolfi, 167, 03100, Frosinone, Italy

2018 – 2021 Cassino, Italy
M.SC. IN MECHANICAL ENGINEERING (LM-33) Università degli Studi di Cassino e del Lazio Meridionale

Website www.unicas.it | **Final grade** With Honours (110/110 cum laude) |

Thesis Preliminary Design of a Daysailer with Hydrofoil, presented and published as Conference Paper in JCM 2024 International Conference, Ischia, 2022.

2022 – 2022 Ithaca, United States
A HANDS-ON INTRODUCTION TO ENGINEERING SIMULATION edX & Cornell University

Mastered fundamentals of **Finite Element Analysis (FEA)** and **Computational Fluid Dynamics (CFD)**.
Applied Ansys software **workflows** for structural and fluid dynamic simulations.

Website https://courses.edx.org/certificates/e9714f7fe96c4e4682302a65cf58b5bc | **Final grade** Professional certification owner

LANGUAGE SKILLS

Mother tongue(s): **ITALIAN**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1
SPANISH	A2	A2	A2	A2	A2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

TECHNICAL EXPERTISE

Technical Skills

Advanced Simulation: CFD, FEA, FSI (Fluid-Structure Interaction) & Topology Optimisation.
Design & Modelling: Algorithmic & Parametric Modelling, Generative Design, DfAM.
Materials & Testing: Composite & Lightweight Structures, Experimental Data Correlation.
R&D Approach: Bio-inspired Design, Reverse Engineering, Rapid Prototyping.
Basic Programming: Python, C (Engineering applications).

Softwares

Numerical Simulation: Simcenter STAR-CCM+, Ansys (Fluent/Mechanical), XFlr5.
CAD & Design: Rhinoceros Grasshopper (Licensed user), Solidworks, Siemens NX.
Computation: MATLAB & Simulink.
General: Microsoft Office, LaTex.

SKILLS

Soft Skills

- **Leadership and Team Guidance** – Led and mentored research and design teams in academic, industrial, and startup contexts, promoting innovation and accountability.
- **Cross-functional Collaboration** – Worked effectively across engineering, research, and production teams, fostering a multidisciplinary approach to problem-solving.
- **Project Planning and Technical Coordination** – Managed timelines, resources, and priorities for complex R&D and prototyping activities, from concept to testing phases.
- **Adaptability and Resilience** – Proven ability to transition between roles (researcher, engineer, instructor), adapting quickly to technical challenges and dynamic environments.
- **Scientific Communication** – Presented technical content at international conferences and to industrial partners; coordinated publication workflows and collaborative papers.
- **Innovative Problem Solving** – Applies lateral thinking to overcome engineering constraints, effectively translating complex theoretical concepts into optimised, practical solutions.

DRIVING LICENCE

Driving Licence: A

Driving Licence: B

Driving Licence: ENAC & EASA Pilot UAS A1-A3

PUBLICATIONS

2026
A Systematic Design Method for Fully Submerged Hydrofoil Wing Systems with Longitudinal Flight Control

Authors: Romolo Di Bernardo, Claudio Pensa, Domenico Speranza & Antonio Gloria | **Journal Name**: Ocean Engineering | **Volume, Issue and Pages**: Vol. 343, Part 3, 123027 | **Publisher**: Springer

2025
A Comparative Study of Reverse Engineering Approaches Toward the Design and Analysis of 2-D High-Speed Hydrofoils

Authors: Romolo Di Bernardo, Antonio Gloria, Stefano Papa & Domenico Speranza | **Journal Name**: Journal of Oceanic Engineering | **Volume, Issue and Pages**: Vol. 50, Issue:3, Pages: 1993-2004 | **Publisher**: IEEE

2025
Visual Programming Approach to T Hydrofoil Design - Conference Paper

International Joint Conference on Mechanics, Design Engineering and Advanced Manufacturing, JCM 2024, Valencia, Spain. June 12-14th, 2024.

Authors: Romolo Di Bernardo, Stefano Papa, Claudio Pensa & Domenico Speranza | **Journal Name**: Lecture Notes in Mechanical Engineering | **Volume, Issue and Pages**: Pages 65 - 73 | **Publisher**: Springer

2025
Preliminary design of a 75 m Mega Yacht with diesel - Electric hybrid propulsion powered with hydrogen FCs

Selected for the Special Issue of Hypothesis International Conference, Muscat, Oman. October 23-25th, 2023.

Authors: Romolo Di Bernardo, Benedetto Di Cecca, Tommaso Coppola, Giuseppe Spazzafumo, Domenico Spernaza | **Journal Name**: International Journal of Hydrogen Energy | **Volume, Issue and Pages**: Vol: 137, Pages: 917 - 924 | **Publisher**: Elsevier

2024
Lift and Drag Coefficients for a NACA 63-412 near the Free Surface - Conference Paper

1st International PhD Symposium on Engineering and Sport Science, Cassino, Italy. October 10 - 12th, 2023.

Authors: Romolo Di Bernardo, Salman Ali, Giuseppe Spazzafumo & Domenico Speranza | **Journal Name**: International PhD Symposium | **Volume, Issue and Pages**: Pages 279 - 283 | **Publisher**: EUC-EDIZIONI UNIVERSITÀ DI CASSINO

2024
Hydrofoil Technology: Current Applications and Future Developments for Sustainable Boating - Conference Paper

3rd International Conference on Design Tools and Methods in Industrial Engineering, ADM 2023, Firenze, Italy.

Authors: Domenico Speranza, Stefano Papa, Claudio Pensa, Romolo Di Bernardo | **Journal Name**: Lecture Notes in Mechanical Engineering | **Volume, Issue and Pages**: Pages 339 - 346 | **Publisher**: Springer

2023
Basic design and virtual prototyping of a hydrofoil hybrid daysailer - Conference Paper

International Joint Conference on Mechanics, Design Engineering and Advanced Manufacturing, JCM 2022, Ischia, Italy.

Authors: Domenico Speranza, Romolo Di Bernardo, Massimo Martorelli, Antonio Gloria, Claudio Pensa & Stefano Papa | **Journal Name**: Lecture Notes in Mechanical Engineering | **Volume, Issue and Pages**: Pages 122 - 134 | **Publisher**: Springer

A complete list of publications is available on my ResearchGate, Google Scholar and Scopus profiles.

CONFERENCES AND SEMINARS

17 MAY 2026 – 20 MAY 2026 Gaeta, Italy.
Organizing Member of PHRS13 – Photosynthesis and Hydrogen Energy Research for Sustainability

14 DEC 2025 – 19 DEC 2025 Sesto (BZ), Italy
ICSC Spoke-3 Final Meeting: The Future of HPC and Big Data in Astrophysics and Cosmology

Presenter of "VIrtual Reality Tools for Understanding scientific ObServatiOns (VIRTUOSO)"
R. Di Bernardo, S. Papa.

8 SEP 2025 – 12 SEP 2025 Instituto Tecnologico de Monterrey, Santiago de Queretaro, Mexico.
HYPOTHESIS (HYdrogen POwer THeoretical & Engineering Solution International Symposium) XX International Conference

Presenter of the paper "A Parametric Design Method of Metal HydrideBulb Keel as Hydrogen Storage System in Sailing Yachts" R. Di Bernardo, C. Pensa, G. Spazzafumo, D. Speranza and A. Gloria. Selected for publish in the *HYPOTHESIS XX* Special Issue on International Journal of Hydrogen Energy (IJHD).

3 SEP 2025 – 5 SEP 2025 University of Genova, Genova, Italy.
ADM 2025 International Conference

Presenter of the conference paper "A Technical Focus on Design and Analysis of 2D High-Speed Hydrofoils" R. Di Bernardo, D. Speranza, S. Papa, M. Martorelli and A. Gloria.

23 OCT 2023 – 26 OCT 2023 Sultan Qaboos University (SQU), Muscat, Oman.

HYPOTHESIS (HYdrogen POwer THeoretical & Engineering Solution International Symposium) XVIII International Conference

Presenter of the paper "Preliminary design of a 75 m Mega Yacht with diesel-Electric hybrid propulsion powered with hydrogen FCs" R. Di Bernardo, B. Di Cecca, T. Coppola, G. Spazzafumo and D. Speranza. Selected for publish in the *HYPOTHESIS XVIII* Special Issue on International Journal of Hydrogen Energy (IJHD).

10 OCT 2023 – 12 OCT 2023 Cassino, Italy.

International PhD Symposium

General Coordinator and Co-Editor for the Conference Proceedings.
Presenter of the paper "Lift and Drag Coefficients for a NACA 63-412 near the Free Surface" R. Di Bernardo, S. Ali, G. Spazzafumo and D. Speranza.

Link https://www.unicas.it/calendario-eventi/international-phd-symposium/

1 JUN 2022 – 3 JUN 2022 Ischia, Italy.

JCM 2022 - INTERNATIONAL JOINT CONFERENCE ON MECHANICS, DESIGN ENGINEERING AND ADVANCED MANUFACTURING

Presenter of the paper "Basic design and virtual prototyping of a hydrofoil hybrid daysailer" D. Speranza, R. Di Bernardo, M. Martorelli, A. Gloria, C. Pensa and S Papa.

A complete list of conferences is available on my ResearchGate, Google Scholar and Scopus profiles.

NETWORKS AND MEMBERSHIPS

IEEE Member, IEEE Oceanic Engineering Society Member

CURRENT

Actually reviewer for International Journals:

- Ocean Engineering (OE) - Elsevier
- Journal of Marine Science and Engineering (JMSE) - MDPI
- Aerospace - MDPI

HOBBIES AND INTERESTS

Watersports enthusiast: active wingfoil and sailing practitioner. Former professional volleyball player and drummer.

I hereby consent to the processing of my personal data in accordance with Article 6.1(a) of the GDPR.

Gaeta , 1 Jan 2026

Romolo Di Bernardo